Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 · STANDARD 6.NOS.4

Find Factors and Multiples

Lesson 6-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.
Language Objective: I can explain my choice using the words factor, multiple, common, greatest, and least.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Factor | Multiple | Greatest common factor (GCF) | Least common multiple (LCM) | Factor pair | Prime factorization | Common factor | Greatest common factor | Least common multiple

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A number that divides another exactly, with no remainder, is a _____.

2. What you land on counting by a number are its _____.

3. The largest number that divides both is their _____ _____ _____.

4. The smallest number both numbers reach is their _____ _____ _____.

5 Two factors that multiply to make the number are a _____ _____.

6 Writing a number as _____ numbers multiplied together is its prime factorization.

7 A factor that two numbers share is a _____.

8 The largest factor two numbers share is the _____.

9 The smallest multiple two numbers share is the _____.

Write About the Math

What is the greatest number of combo packs the manager can make, and what goes in each one?
¿Cuál es el mayor número de paquetes combo que puede hacer el gerente, y qué lleva cada uno?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Factor**
  Factor
- ☐ **Multiple**
  Múltiplo
- ☐ **Greatest common factor (GCF)**
  Máximo común divisor (MCD)
- ☐ **Least common multiple (LCM)**
  Mínimo común múltiplo (mcm)
- ☐ **Factor pair**
  Par de factores

Model: *The number of pods must divide BOTH 24 and 36 evenly.* — Students see the number of pods has to be a common factor of 24 and 36, so it must divide both with nothing left over, ruling out numbers that only divide one of them.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The number of packs must be a ____ of both 48 and 36. Their common factors are 1, 2, 3, 4, 6, and ____. The greatest of these is the ____, so the manager makes ____ packs, each holding ____ pencils and 3 notebooks.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.
Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Factor
2. **Notes 2:** Multiple
3. **Notes 3:** Greatest common factor (GCF)
4. **Notes 4:** Least common multiple (LCM)
5. **Notes 5:** Factor pair
6. **Notes 6:** Prime factorization
7. **Notes 7:** Common factor
8. **Notes 8:** Greatest common factor
9. **Notes 9:** Least common multiple
10. **Watch for this mistake:** *A common mistake is reaching for the wrong tool: listing FACTORS for a problem about repeating cycles. Ayasha's classes repeat every 6 and 8 weeks, so the weeks they land on are multiples — answering with the GCF of 2 claims a class that meets every 8 weeks happens in week 2. Use the size of the answer as a check: a GCF is never larger than the smaller number, and an LCM is never smaller than the larger one.*
11. **Try It 1:** *— When two numbers share no factor except 1, their GCF is 1 — which is a real answer, not a mistake.*
12. **Try It 2:** *— Factors are finite and sit at or below the number; multiples are infinite and sit at or above it. Knowing this checks your answer before you write it down.*